

Tensor computations

Border Comon's conjecture over the complex numbers

Difficulty: extreme **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Extreme because equality of two secant-rank notions for all symmetric tensors remains a foundational tensor-geometry barrier; community importance concerns whether symmetry can be imposed without an approximation-rank penalty.

Statement

For every integer order $d \geq 3$, dimension $n \geq 2$, and symmetric tensor $T \in \text{Sym}^d(\mathbb{C}^n)$, is

$$\underline{R}(T) = \underline{R}_{\text{sym}}(T)?$$

Here $R(T)$ is the least number of summands in a decomposition $T = \sum_{j=1}^r v_{j,1} \otimes \cdots \otimes v_{j,d}$, with complex vectors. Symmetric rank restricts each summand to $c_j v_j^{\otimes d}$. The border rank $\underline{R}(T)$ is the least r for which a sequence of tensors of rank at most r converges to T in the entrywise Euclidean topology. Symmetric border rank instead uses symmetric tensors of symmetric rank at most r .

Relevance

This asks whether exploiting symmetry can increase the number of rank-one terms required for arbitrarily accurate approximation. It concerns the attainable accuracy of symmetric low-rank tensor models.

References

1. T. Mańdziuk and E. Ventura, *Symmetrization maps and minimal border rank Comon's conjecture*, arXiv:2411.05721v2, 2026. [Full text](#), §1, opening discussion and Conjecture 1.1; the general conjecture precedes the minimal-border-rank specialization. [Version record](#).
2. J. M. Landsberg, *Geometry and Complexity Theory*, Cambridge University Press, 2017. [Author text](#), Conjecture 5.6.1.5, a three-factor formulation.

Status check — 2026-09-10

Rechecked [Mańdziuk–Ventura v2, §1 and the main theorems](#), and searched for 2025–2026 proofs or counterexamples. The September 2026 revision explicitly leaves the general border conjecture open while proving minimal-border-rank subfamilies. Those are genuine portions of this universal target; Shitov's counterexample to ordinary exact-rank Comon is a different statement. No full border-rank resolution was located.